6 Discussion

[To be completed: Write the discussion yourselves. Evidence available in results/tables/ and docs/summary.md: (1) share of DY connectedness explained by the common factor (41%, stable across robustness) and its link with the time pattern; (2) how the method changes the picture (density, agreement, stability, Table 2); (3) interpretation of net transmitters/receivers, including Argentina (role in the paper still to be defined by the group); (4) limitations: returns instead of volatilities, ETF/currency effects, short windows for FNETS and NETS (numerical safeguards, tuning-driven drops), sparse methods leave almost no lagged links; (5) the team's database-based extension (to be added once the code is shared).]

7 Conclusion

[To be completed: Answer the research question and state what can be claimed with confidence. Final wording of the question and hypotheses is still to be decided by the group.]

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9 Data and Code Availability

The data (weekly ETF returns) and the full code are available at `https://github.com/ezellabra/ML4E-code`. The pipeline is written in Python; NETS and FNETS are estimated in R (packages `nets` 0.9.1 and `fnets` 0.1.6) through a command-line bridge. All parameters are in `config.yaml` and the whole analysis is reproduced with `make all` (or `python run_all.py`).

A Robustness

Table 3: Robustness: mean total connectedness (%) by method, share of DY connectedness lost once the common factor is removed (FNETS), and FNETS common-variance share.

Scenario	Windows	DY	Lasso VAR	NETS	FNETS (idio.)	Lost w/o factor (%)	Common var. (%)
Baseline (w=104, H=4)	179	81.0	80.0	61.5	47.2	41	58
H=2	179	80.9	80.0	61.5	47.2	41	58
H=8	179	81.0	80.1	61.5	47.2	41	58
w=78	185	81.2	79.6	57.8	50.5	38	58
w=130	172	81.1	80.5	63.4	46.2	43	58
Excluding SPY	179	80.0	78.9	59.9	45.9	42	58
Excluding windows touching 2020	141	79.7	78.6	58.8	47.2	41	55

Table 4: Static connectedness: one estimate on the whole period (full sample) and on the subsamples before and after 2020. These are single long-sample estimates, not averages over rolling windows, so they differ from Tables 2 and 3.

Sample	Method	Total conn. (%)	LR p.corr. dens. (%)
Full sample	DY	81.9	100.0
Full sample	Lasso VAR	82.1	100.0
Full sample	NETS	73.5	60.1
Full sample	FNETS (idio.)	34.2	0.0
2011-2019	DY	83.0	100.0
2011-2019	Lasso VAR	82.8	100.0
2011-2019	NETS	69.6	60.1
2011-2019	FNETS (idio.)	34.1	2.0
2021-2026	DY	73.5	100.0
2021-2026	Lasso VAR	73.7	100.0
2021-2026	NETS	56.0	52.3
2021-2026	FNETS (idio.)	39.6	13.7

Table 3 reruns every method under alternative choices. The levels of Diebold–Yilmaz and lasso connectedness hardly move: between 79.7% and 81.2% and between 78.6% and 80.5% across all scenarios. NETS is the most sensitive to the window length (57.8% with 78-week windows, 63.4% with 130-week windows), as shorter windows leave the penalized estimator less information and make it keep fewer links. FNETS idiosyncratic connectedness ranges from 45.9% to 50.5%. The horizon matters very little: Diebold–Yilmaz changes by about 0.1 points between